INCH-POUND

MIL-STD-1861A

28 March 1993

SUPERSEDING

MIL-STD-1861

23 August 1989

MILITARY STANDARD

ELECTRICAL AND ELECTRONIC ASSEMBLIES, BOARDS, CARDS, AND ASSOCIATED HARDWARE, SELECTION AND USE OF

AMSC N/A

FSC 5998

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FOREWORD

1. This military standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Research Laboratory, ATTN: AMSRL-EP-RD, Fort Monmouth, NJ 07703-5601 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

3. This standard provides a list of electrical and electronic assemblies, boards, cards, and associated hardware for use in the design of military equipment. The application information and performance characteristics contained in this standard are offered for guidance and are not to be considered as binding. Additions to this standard will be made periodically as the referenced documents are updated.

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1 SCOPE

1 1 Scope This standard establishes the criteria for the design and treatment of printed wiring boards, circuit card assemblies, and associated hardware

1 2 Purpose The purpose of this standard is to

a Provide the designers of new equipment with a selection of standard items for military applications

b Control and minimize the variety of items used in military equipment in order to facilitate logistic support of the equipment in the field

c Outline criteria pertaining to the use of items for design purposes for new equipment

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2 APPLICABLE DOCUMENTS

2 1 Government documents

2 1 1 Specifications, standards, and handbooks The following specifications, standards, and handbooks form a part of this document to the extent specified herein Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications (DODISS) and supplement thereto, cited in the solicitation

SPECIFICATIONS

MILITARY

MIL-S-13949 - Sheet, Printed Wiring Board, General Specification for.
MIL-A-24741 - Assemblies, Metal Electrical Backplane, One and Two Layer, General Specification for
MIL-C-28809 - Circuit Card Assemblies, Rigid, Flexible, and Rigid-Flex
MIL-A-28870 - Assemblies, Electrical Backplane, Printed Wiring, General Specification for
MIL-P-50884 - Printed Wiring, Flexible, and Rigid-Flex
MIL-P-55110 - Printed Wiring Board, General Specification for

STANDARDS

MILITARY

MIL-STD-275 - Printed Wiring for Electronic Equipment
MIL-STD-2118 - Flexible and Rigid-Flex Wiring for Electronic Equipment, Design Requirements for.
MIL-STD-2119 - Design Requirements for Printed Wiring Electrical Backplane Assemblies
MIL-STD-2198 - Design Requirements for Metal Elertrical Backplane Assemblies

HANDBOOKS

FEDERAL

H-6 - Alphabetic Index of Names
H2-2 - Cataloging Handbook

(Copies of federal and military specifications and standards are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094 Not more than five items may be ordered on a single request, the invitation for bid or contract number should be cited where applicable Only the latest revisions (complete with latest amendments) are available, slash sheets must be individually requested Request all items by document number For information on subscription service, direct inquiries to the above address or telephone (215) 697-3321, inquiry desk Copies of Cataloging Handbooks H2-2 and H6 are available from the Commander, Defense Logistics Services Center, Battle Creek, MI 49017-3084 Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Naval Publications and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093)

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2 1.2 Other Government documents, drawings, and publications The following other Government documents, drawings, and publications form a part of this document to the extent specified herein Unless otherwise specified, the issues are those cited in the solicitation

DRAWINGS

83023	-	Extractor, Electrical Card, Plastic
84006	-	Electrical Card Holders
84101	-	Electrical Card Holders
84103	-	Electrical Card Holders, Retainers
84168	-	Electrical Card Holders, Retainers
84191	-	Extractor, Electrical Card, Metal
85020	-	Holder, Electrical Card.
85021	-	Holder, Electrical Card
85033	-	Holder, Electrical Card
85034	-	Holder, Electrical Card
85067	-	Electrical Card Holders
85068	-	Holder, Electrical Card.
89029	-	Holder, Electrical Card, Anti-vibration
89064	-	Holder, Electrical Card, Retainer, 5-Piece Wedge Lock

(Copies of DESC drawings are available from the Defense Electronics Supply Center, ATTN DESC-ESP, 1507 Wilmington Pike, Dayton, OH 45444-5099 Only the latest revisions (complete with latest Notice of Revisions) are available Request all items by document number For information on concerning requesting these drawing, direct inquiries to the above address or telephone (513) 296-6095 or DSN 986-6095)

2 2 Non-Government publications The following documents form a part of this document to the extent specified herein Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation

INSTITUTE FOR INTERCONNECTING AND PACKAGING ELECTRONIC CIRCUITS (IPC)

IPC-DW-425/11 - Design and End Product Requirements for Discrete Wiring Boards/Plated-Through Hole Connection

(Application for copies should be addressed to the Institute for Interconnecting and Packaging Electronic Circuits, 7380 North Lincoln Avenue, Lincolnwood, IL 60646)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents These documents also may be available in or through libraries or other informational services)

2 3 Order of precedence In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained

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3 DEFINITIONS

3 1 Definitions of item names (H-6) Definitions of item names are in the Federal Item Name Directory for Supply Cataloging, H 6, section A

3 2 List of item names (H2-2) A complete list of item names in FSC 5998 can be found in Cataloging Handbook H2-2

3 3 Definitions The meanings of terms used in this standard are in accordance with military specifications and DESC drawings

3 3 1 DESC drawing A standardization document developed by and available from the Defense Electronics Supply Center

3 3 2 Part or Identifying number (PIN) Formerly part number (p/n)

3 3 3 Qualified Products List (QPL) A document maintained by the government qualifying activity which lists vendors whose products are officially approved by the Government

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4 GENERAL REQUIREMENTS

4 1 Item identification PIN's and type designations are used to identify the items listed in this standard, and shall be as specified in the individual specifications

4 2 Conflict of requirements In the event of conflict between the technical requirements of items described in this standard and the applicable specification or DESC drawing, the specification or DESC drawing shall govern

4 3 Criteria for inclusion in this standard The criteria for inclusion in this standard are as follows

a The item(s) are classified in FSC 5998

b The item(s) shall be the best type available for general use in military equipment

c The item(s) are covered by military specifications or DESC drawings

d The item(s) shall be available from at least one source

e The item(s) shall be used for design purposes in new equipment

4 4 Order of preference The order of preference shall be as follows

a Military specification parts

b Military drawing parts

c Other parts

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5 DETAILED REQUIREMENTS

5 1 Printed wiring board size Whenever cost and technical requirements permit, preferred rigid printed wiring board sizes should be used These printed wiring board sizes will facilitate the development and use of standardized insertion end extraction tools The preferred printed wiring board sizes, extractor hole size and location are shown on figure 1

5 2 Rigid printed wiring and printed wiring boards Rigid printed wiring and printed wiring boards for single-sided, double-sided, and multilayer printed wiring shall conform to MIL-STD-275 and MIL-P-55110 The base materials and prepreg used for rigid printed wiring boards shall conform to MIL-S-13949

5 3 Rigid circuit card assemblies Rigid circuit card assemblies consisting of rigid printed wiring boards on which separately manufactured parts have been added shall conform to MIL-C-28809

5 4 Flexible and rigid-flex wiring Flexible and rigid-flex printed wiring shall conform to MIL-P-50884 and shall be designed in accordance with MIL-STD-2118

5 5 Flexible and rigid-flex circuit card assemblies Flexible and rigid-flex circuit card assemblies consisting of flexible or rigid-flex printed wiring boards on which separately manufactured parts have been added shall conform to MIL-C-28809

5 6 Discrete wiring boards Discrete wiring boards with plated-through holes shall be in accordance with IPC-DW-425/11

5 7 Backplane assemblies, printed wiring Electrical backplane printed wiring assemblies shall conform to MIL-A-28870 and shall be designed in accordance with MIL-STD-2119

5 8 Metal electrical backplane assemblies Metal electrical backplane assemblies shall conform to MIL-A-24741 and shall be designed in accordance with MIL-STD-2198

5 9 Associated hardware

5 9 1 Holders Circuit card holders and guides shall be as specified in tables I through VII for metals, and tables VIII through X for plastics

5 9 2 Extractors and retainer-ejectors Extractors and retainer-ejectors shall be as specified in tables XI and XII

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TABLE I Holders, electrical card, retainer, metal, 3 wedge

Mounting length (inches)	Number of mounting holes	Mounting method	DESC drawing PIN
2 8	2	Rivets	84103-01, 08, 15, 18, 25, 32, 38
3 0	"	"	84103-02, 09, 19, 26, 39, 45
3 5	"	"	84103-03, 10, 20, 27, 40, 46
3 8	"	"	84103-04, 11, 16, 21, 28, 33, 41
4 0	"	"	84103-47
4 5	"	"	84103-48
4 8	"	"	84103-05, 12, 17, 22, 29, 34, 42
5 0	"	"	84103-49
5 3	"	"	84103-06, 13, 23, 30, 43
5 8	"	"	84103-07, 14, 24, 31, 44
6 1	"	"	84103-36
6 7	"	"	84103-37

TABLE II Holders, electrical card, retainer, metal, 5 wedge

Mounting length (inches)	Number of mounting holes	Mounting method	DESC drawing PIN
2 8	2	Rivets or screws	89029-01, 11
3 0	"	"	89029-02, 12
3 5	"	"	89029-03, 13
3 8	"	"	89029-04, 14
4 0	"	"	89029-05, 15
4 5	"	"	89029-06, 16
4 8	"	"	89029-07, 17, 89064-01 to 12
5 0	"	"	89029-08, 18
5 5	"	"	89029-09, 19
5 8	"	"	89029-10, 20, 89064-13 to 24

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TABLE III. Electrical card holders, metal, slot width .0312 inch

Overall length (inches)	Number of mounting holes	DESC drawing PIN
2.0	2	85020-002
2.5	"	85020-026
3.0	"	85020-050
3.5	"	85020-074
4.0	"	85020-098
4.5	"	85020-122
5.0	"	85020-146
5.5	"	85020-170
6.0	"	85020-194

TABLE IV Electrical card holders, metal, slot width .0625 inch

Overall length (inches)	Number of mounting holes	DESC drawing PIN
2.0	2	85020-005
2.5	"	85034-01
2.50	"	85020-029
2.50	"	85021-002
3.0	2	84168-002
3.0	"	85020-053
3.0	"	85021-017
3.25	3	85034-03
3.5	2	84168-010
3.5	"	85020-077
3.5	"	85021-032
4.0	2	84168-018
4.0	"	85020-101
4.0	"	85021-047
4.25	4	85034-05
4.5	2	84168-026
4.5	2	85020-125
4.5	3	85033-001
5.0	3	84168-034
5.0	2	85020-149
5.0	2	85021-062
5.25	5	85034-07
5.5	3	84168-042
5.5	2	85020-173
6.0	3	84168-050
6.0	2	85020-197
6.0	2	85021-077
6.0	4	85033-017
6.25	6	85034-09
7.0	3	84168-058
7.5	3	84168-066
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]

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TABLE V Electrical card holders, metal, slot width .0937 inch

Overall length (inches)	Mounting method	DESC drawing PIN
1.5	2 holes	85033-135
2.0	"	85020-008
2.25	"	85034-02
2.50	"	85020-032
2.50	"	85021-005
3.00	2 holes	85020-056
3.00	"	85021-020
3.00	"	84168-004
3.00	3 holes	85033-147
3.25	3 holes	85034-04
3.5	2 holes	85020-080
3.5	"	85021-035
3.5	"	84168-012
4.0	2	85020-104
4.0	"	85021-050
4.0	"	84168-020
4.25	4	85034-06
4.5	2	84168-028
4.5	2	85020-128
4.5	3	85033-005
5.0	2	85020-152
5.0	2	85021-065
5.0	3	84168-036
5.25	5	85034-08
5.5	2	85020-176
5.5	3	84168-044
6.0	4	85033-020
6.0	2	85020-200
6.0	2	85021-080
6.0	3	84168-052
6.25	6	85034-10
7.0	3	84168-060
7.5	3	84168-068
8.25	8	85034-12
8.5	3	84168-076

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TABLE VI Electrical card holders, metal, slot width .125 inch

Overall length (inches)	Mounting holes	DESC drawing PIN
1.5	2	85033-185
2.0	"	85033-189
2.5	"	85033-031
3.0	3	85033-035
3.5	"	85033-039
4.0	"	85033-043
4.5	"	85033-047
5.0	4	85033-051
5.5	"	85033-054
6.0	"	85033-057

TABLE VII Electrical card holders, metal, slot width .063 inch

Overall length (inches)	Number of mounting holes	DESC drawing PIN
2.0	2	85068-01
3.5	"	85068-02
5.0	"	85068-03

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TABLE VIII Electrical card holders, plastic, slot width .078 inch.

Overall length (inches)	Number of snap pylons	DESC drawing PIN
2.5	2	84006-01
3.0	"	84006-02
3.5	"	84006-03
4.0	"	84006-04
4.5	"	84006-05
5.0	"	84006-06
5.5	"	84006-07
6.0	"	84006-08
6.5	"	84006-09
7.0	"	84006-10
7.5	"	84006-11
8.0	"	84006-12
8.5	3	84006-13
9.0	3	84006-14
9.5	"	84006-15
10.0	"	84006-16
10.5	"	84006-17
11.0	"	84006-18
11.5	"	84006-19
12.0	"	84006-20
12.5	"	84006-21
13.0	"	84006-22
13.5	"	84006-23
14.0	"	84006-24

TABLE IX Electrical card holders, plastic, deep slot.

Overall length (inches)	Number of snap pylons	DESC drawing PIN
2.50	2	84101-01
4.50	"	84101-02
6.00	"	84101-03
8.00	3	84101-04
4.50	2	84101-22
6.00	2	84101-23
8.00	3	84101-24

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TABLE X Electrical card holders, plastic, slot width .050 to .100 inch

Overall length (inches)	Number of snap pylons	DESC drawing PIN
2.50	2	85067-08, 09, 12
3.50	"	84006-07, 11
4.00	"	89029-01
4.125	"	85067-05, 06, 10
5.50	"	85067-04
6.00		89029-02
6.375	"	85067-03
6.875	"	85067-02
7.00	"	85067-01

TABLE XI Extractor, electrical card, plastic

Material	Slot width (inches)	DESC drawing PIN
Nylon	.075	83023-01*
Nylon	.075	83023-02*
Nylon	.075	83023-03*
Nylon	.075	83023-04*
Nylon	.105	83023-05*
Nylon	.105	83023-06*
Nylon	.105	83023-07*
Nylon	.100	83023-08*
Glass filled nylon	.075	83023-09*
Glass filled nylon	.140	83023-10*
Glass filled nylon	.075	83023-11*
Glass filled nylon	.075	83023-12*
Nylon	.070	83023-13*
Glass filled polycarbonate	.075	83023-14*
Nylon	.070	83023-15*
Nylon	.100	83023-16*
Glass filled polycarbonate	.140	83023-17*
Nylon	.070	83023-18*
Nylon	.075	83023-19*
Nylon	.100	83023-20*
Nylon	.140	83023-21*
Nylon	.075	83023-22*
Nylon	.075	83023-23*

*Indicates color

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TABLE XII Extractor, electrical card, metal

Slot inch	Configuration	DESC drawing PIN
040		84191-01*#
040		84191-02*#
075		84191-03*#
075	Angled handle,	84191-04*#
105	no lock tab	84191-05*#
105		84191-06*#
142		84191-07*#
142		84191-08*#
040		84191-09*#
040		84191-10*#
075		84191-11*#
075	Angled handle,	84191-12*#
105	with lock tab	84191-13*#
105		84191-14*#
142		84191-15*#
142		84191-16*#
040		84191-17*#
040		84191-18*#
075		84191-19*#
075	Straight handle,	84191-20*#
105	with lock tab	84191-21*#
105		84191-22*#
142		84191-23*#
142		84191-24*#
040		84191-25*#
040		84191-26*#
075		84191-27*#
075	Straight handle,	84191-28*#
105	no lock tab	84191-29*#
105		84191-30*#
142		84191-31*#
142		84191-32*#

* Indicates finish The finish options are black anodize "B", chemical film "C", chemical film with the electronic sensitive device type "E" and passivation "P" The "E" finish is preferred for Air Force use

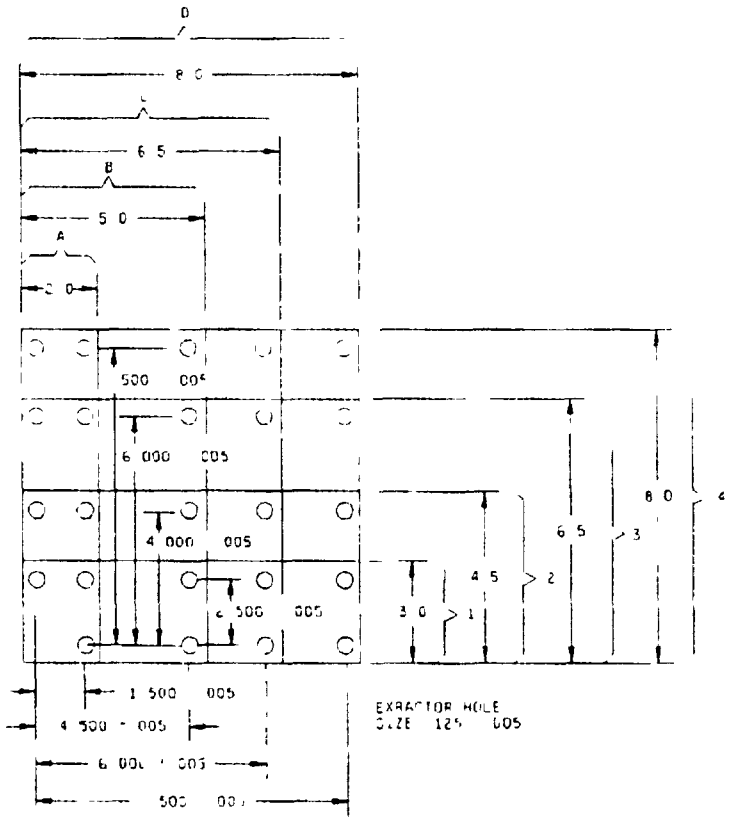
\# Indicates material The material options are 0 063 inch thick aluminum alloy 5052-H32 "A", Type 301 stainless steel "S", and 0 063 inch thick aluminum alloy 6061-T6

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Preferred thicknesses	
Thickness	Tolerance
062	± 001
093	± 0015
125	± 002

Inches	mm
001	0 03
0015	0 038
002	0 07
005	0 13
015	0 38
062	1 57
093	2 36
125	3 18
1 500	38 100
2 0	51
2 500	63 50
3 0	76
4 000	101 60
4 500	114 30
5 0	127
6 000	152 40
6 5	165
7 500	190 50
8 0	203

Board number	Printed wiring board size in nominal ± 015
A1	2 0 x 3 0
A2	2 0 x 4 5
A3	2 0 x 6 5
A4	2 0 x 8 0
B1	5 0 x 3 0
B2	5 0 x 4 5
B3	5 0 x 6 5
B4	5 0 x 8 0
C1	6 5 x 3 0
C2	6 5 x 4 5
C3	6 5 x 6 5
C4	6 5 x 8 0
D1	8 0 x 3 0
D2	8 0 x 4 5
D3	8 0 x 6 5
D4	8 0 x 8 0



NOTES
1 Dimensions are in inches
2 Metric equivalents are for general information only

FIGURE 1 Preferred printed wiring board size

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6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Documents conforming to the requirements of this standard are intended for the use as military standardization documents and are listed in the DODISS. The purpose of this standard is to standardize the preparation of military standards, handbooks, drawings, and bulletins to ensure the inclusion of data and descriptions essential to the selection and application of items and processes, and to aid in the use and analysis of DoD standardization documents.

6.2 Subject term (key word) listings.

Backplane assembly
Circuit card assembly
Discrete wiring board
Extractor, electrical card
Holder, electrical card
Printed circuit board
Printed wiring board
Retainer-ejector, electrical card

CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - ME
Air Force - 17, 99
DLA - ES

Preparing activity
Army - ER

Agent
DLA - ES

(Project 5998-0031)

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